Lecture 03
Entity Relationship Modelling

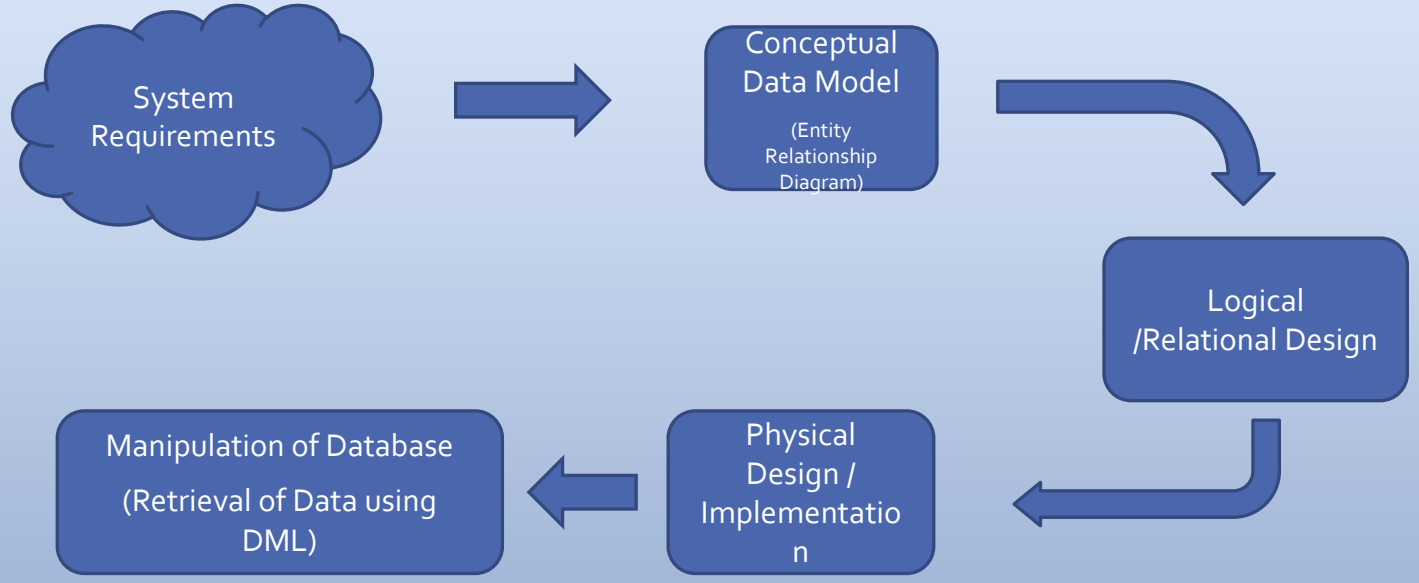
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Lecture Objectives

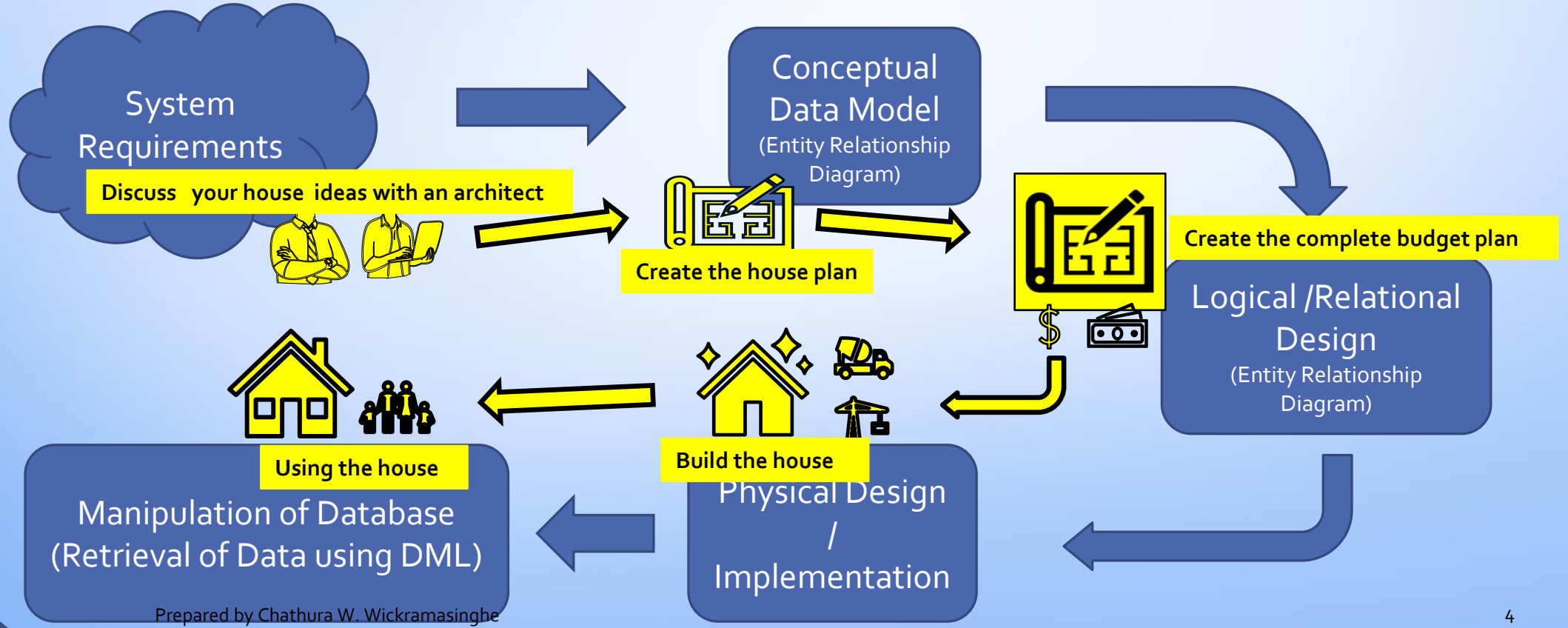
- Understand the different phases in designing the databases
- Conceptual Design
- ERD Basics

Steps in Implementing the Database

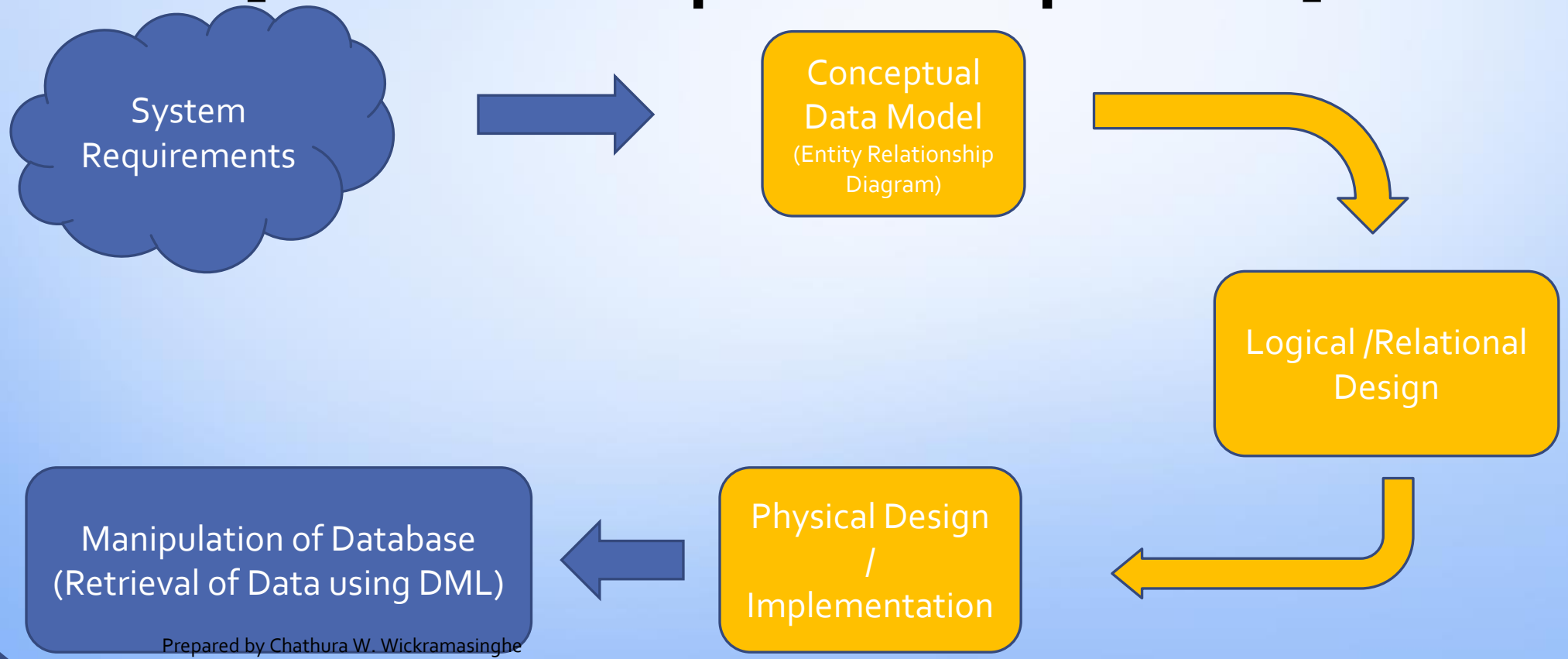


Steps in Implementing the Database

[Explained ..(Steps compared with building a house..)]



Steps in Implementing the Database
[Three main phases explained]



System Requirements for the Database Design

Determine	Determine the purpose of your database.
Find and organize	Find and organize the information required.
Gather	Gather all the types of information you might want to record in the database. Types of data are as follows: •People •Places •Things

Conceptual Database Design

The process of constructing a model of the data used in an enterprise, independent of all physical considerations.

The purpose of the conceptual design phase is to build a conceptual model based upon the previously identified requirements, but closer to the final physical model. A commonly-used conceptual model is called an entity-relationship model.



Why the conceptual design phase?

[Make your own notes..]

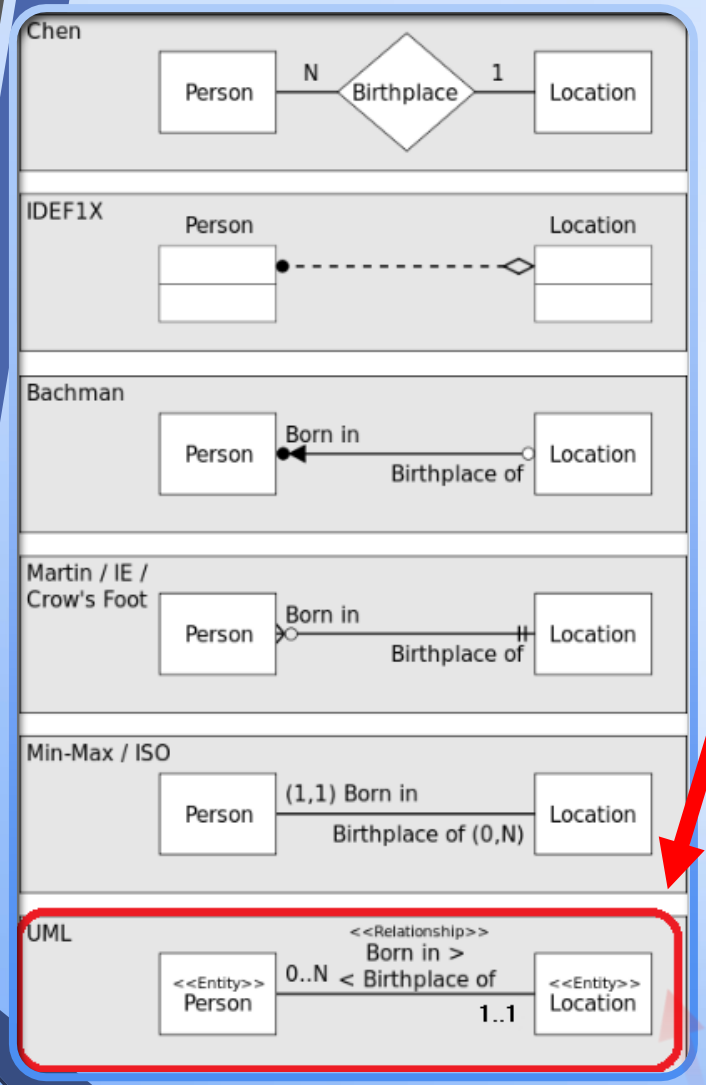
Conceptual Design **[Explained..]**

The purpose of the conceptual design phase is to build a conceptual model based upon the identified requirements, but closer to the final physical model.

A commonly-used conceptual model is called an **entity-relationship model.**

Entity Relationship Model

- The purpose of the conceptual design phase is to build a conceptual model based upon the identified requirements, but closer to the final physical model.
- A commonly-used conceptual model is called an entity-relationship model.
- The entity-relationship model (ERM) is a conceptual model that represents the information structure of a problem domain in terms of entities and relationships.
- The result of modeling using the ERM is graphically represented as an entity relationship diagram (ERD).
- The ERM was introduced by Peter Chen in 1976.

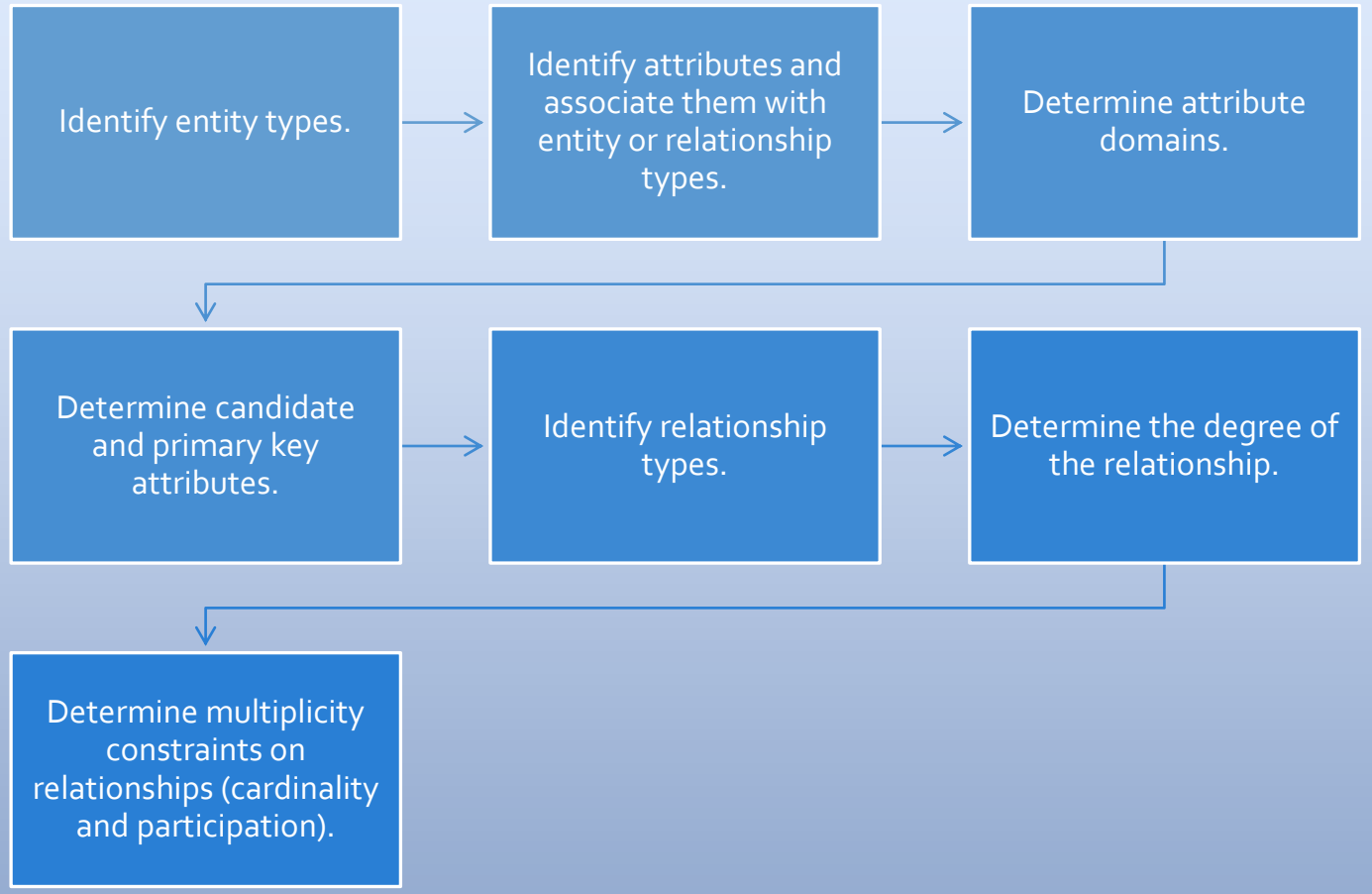


Different ERD Notations Used in the Industry.

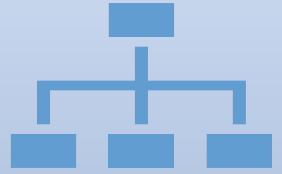
- In this module you will learn to implement ERD using the UML notation. (In the coursework and exams, you will be awarded marks only for the usage of UML notations to implement ERD.

- You can use the free tool given below to draw the ER Diagrams online using the UML notation.
  - https://app.creately.com/

Steps in Creating the Entity Relationship Diagram



What is an Entity Type and Entity?



What is an Entity Type?

Group of objects with the same properties, identified by the enterprise as having an independent existence.

These objects may be tangible (concrete/physical) or intangible (conceptual/abstract).

Entity is named using a noun or noun clause.

Entity name should be in the singular form.

Often referred to simply as an entity.



What is an ENTITY?

AKA INSTANCE OF AN ENTITY TYPE.

Uniquely identifiable object or instance of an entity type.

Examples of Different Entity Types

- **Entity Types with Physical Existence**
  - Staff
  - Property
  - Customer
  - Part
  - Supplier
  - Product
- **Entity Types with Conceptual Existence**
  - Inspection
  - Preferences
  - Inspection
  - Sale
  - Work Experience

Rules to be satisfied to become an ENTITY.

- Each occurrence should be uniquely identifiable.
- There must be more than one occurrence of the entity.
- The entity must have data (more than one attribute) that is required to be stored.
- It must be of direct interest to the system under consideration.

UML Conventions

The first letter of each word in the entity name is uppercase and the rest of the letters are lowercase, and all words are joined together.

E.g. - Branch, Staff, PropertyForRent, Student, Course, VideoCopy, Reservation

Attributes

- What is an attribute?
  - Property of an entity or a relationship type. E.g. – the EMPLOYEE entity may have attributes name, address, date of birth, salary, and gender.
- What is an attribute domain?
  - Set of allowable values for one or more attributes.

Attribute Classifications

Based on the complexity of attributes

- Simple or composite

Based on the Values

- Single Valued, Multi-Valued, or Derived.

Simple Value Attributes

- Attribute composed of a single component with an independent existence.
  - E.g. – name, position, salary

Composite Attributes

- Attribute composed of multiple components (simple attributes), each with an independent existence.
  - E.g., address (street, city, postcode)

Single-valued Attribute:

- Attribute that holds a single value for an entity occurrence.

Multi-valued Attribute:

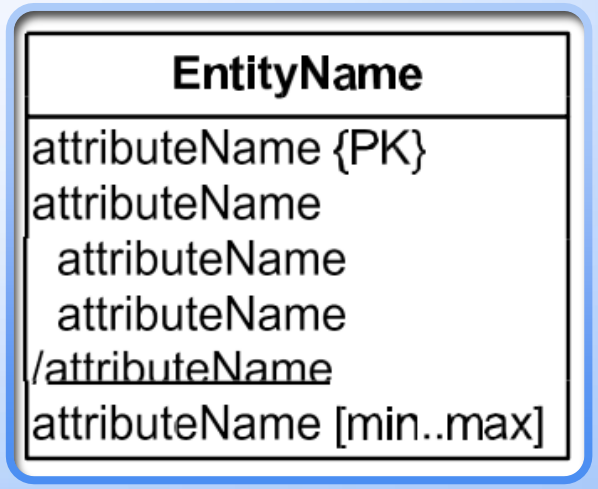
- Attribute that holds multiple values for an entity occurrence.
  - E.g., telNo [1..3] denotes up to 3 telephone numbers.

Derived Attribute:

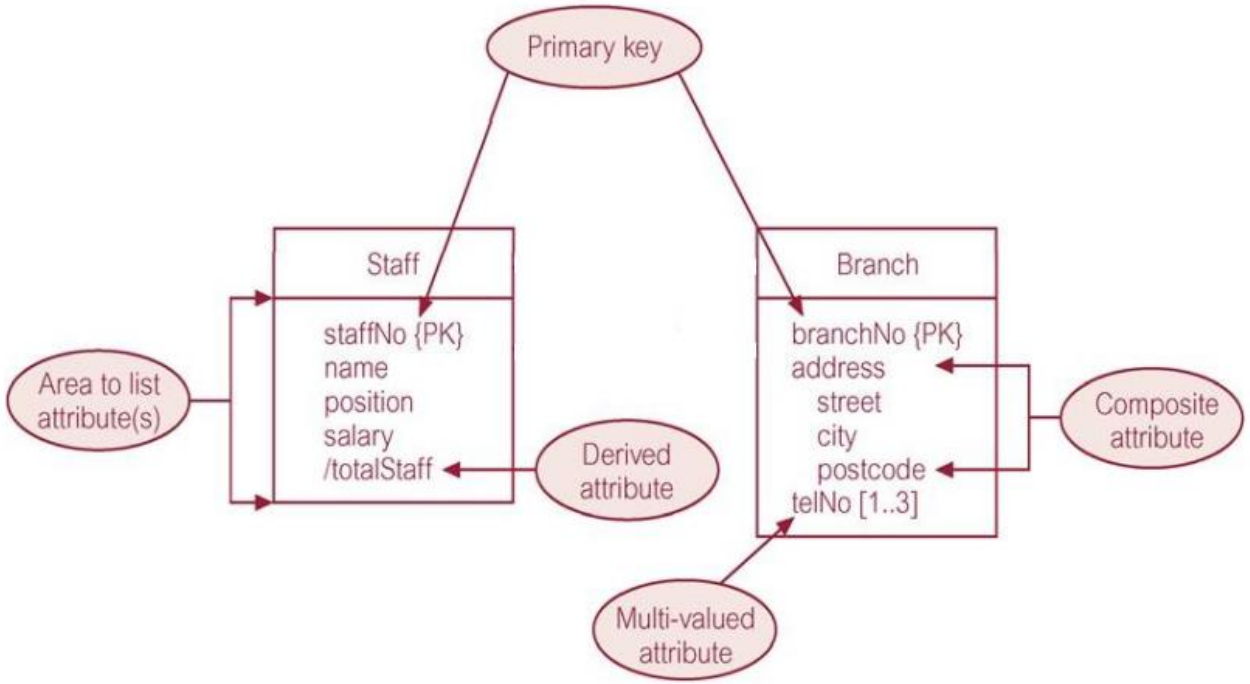
- Attribute that represents a value that is derivable from the value of a related attribute, or set of attributes, not necessarily in the same entity type.
  - E.g., /total staff
- Derived attributes are preceded with a / (forward slash)

Attributes UML Conventions

- If attributes for entity type are to be shown, divide the rectangle representing the entity into two parts.
- First list the primary key attribute, labeled with {PK}.
- The first letter in the attribute name is lowercase and the first letter of each subsequent word is uppercase.
- Multi-Valued Attributes - E.g., telNo [1..3] denotes up to 3 telephone numbers.
- Derived attributes are preceded with a / (forward slash) E.g., /total staff



Entity Attributes Example



Refer to below case study and find out the entities and the attributes.

- The university offers one or more programs. A program is made up of one or more courses. A student must enroll in a program. A student takes the courses that are part of her program.
- A program has a name, a unique program identifier, the total credit points required to graduate, and the year it commenced.
- A course has a name, a unique course code, a credit point value, and the year it commenced.
- Students have a name(First name, Last name), a unique ID, a date of birth, age and several contact numbers.

References and Further Reading

- T CONNOLLY & C BEGG. Database Systems – A Practical Approach to Design, Implementation, and Management, 4th Edition. Addison-Wesley,2005.
  - Chapter 11 – Entity-Relationship Modelling.
- • T CONNOLLY & C BEGG. Database Solutions – A step- by step guide to building databases, 2nd Edition. Addison-Wesley, 2004.
  - Chapter on Entity-Relationship Modelling.